

Lunar Surface Infrastructure Program WBS

Work Breakdown Structure Level 1 and Level 2 scope architecture version 0.1

Owner	Program Director	Status	Draft for review
Version	0.1	Basis	Program Charter v1.1

Purpose and control objective

This WBS organizes the complete program scope into mutually understandable Level 2 elements. It establishes the structure for requirements allocation, scheduling, resource planning, risk ownership, performance measurement, and configuration control. It does not yet assign dates, budgets, control-account managers, or contractual authority.

Program scope architecture

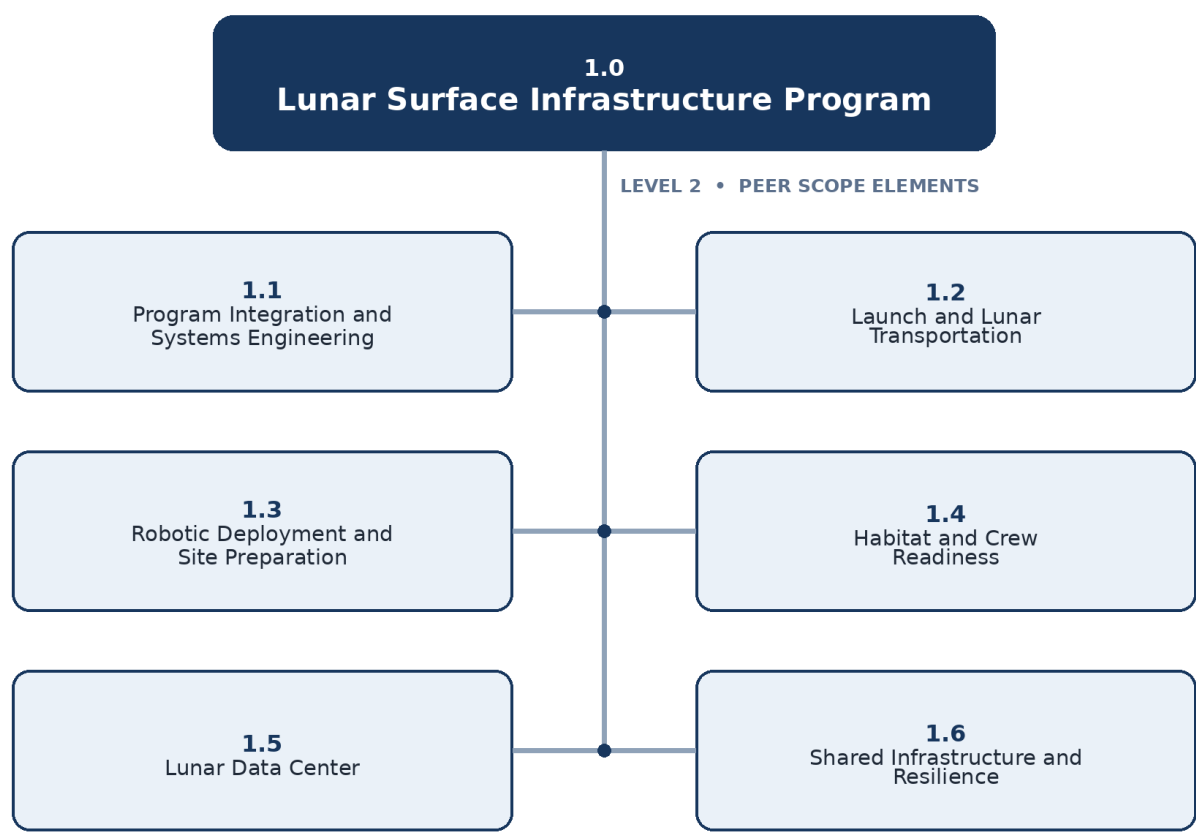


Figure 1. Level 1 program scope decomposed into six peer Level 2 elements.

WBS hierarchy

WBS	Level	Element	Primary deliverable
1.0	1	Lunar Surface Infrastructure Program	Integrated lunar demonstration planning baseline.
1.1	2	Program Integration and Systems Engineering	Controlled program baseline and integrated decision framework.
1.2	2	Launch and Lunar Transportation	Robotic, habitat, utility, and computing assets delivered to the lunar surface.
1.3	2	Robotic Deployment and Site Preparation	Surveyed, prepared, equipped, and remotely verified operating site.
1.4	2	Habitat and Crew Readiness	Operational habitat capable of supporting seven crew members.
1.5	2	Lunar Data Center	Commissioned computing capability within approved operating limits.
1.6	2	Shared Infrastructure and Resilience	Common utilities and recovery capabilities supporting all projects.

Decomposition rules

- The WBS is deliverable-oriented: each element represents a defined program outcome rather than an organization or calendar phase.
- Project-specific equipment and work remain within the applicable project element; services supporting multiple projects belong under 1.6.
- Program-level governance, integration, requirements, interfaces, configuration control, and readiness evidence belong under 1.1.
- Each lower-level element must map to one Level 2 parent and must not duplicate scope assigned elsewhere.
- Schedule activities, budgets, resources, risks, and performance measures will be assigned only after the WBS and dictionary are approved.

Baseline decision

Approve the Level 1 and Level 2 structure as the scope architecture for development of the WBS Dictionary and Level 3 deliverables.

Level 2 Element Definitions

Scope boundaries and completion conditions

1.1 Program Integration and Systems Engineering

Includes program governance, requirements and interface management, integrated schedule and resource planning, configuration control, risk and opportunity management, technical integration, verification planning, readiness reviews, and executive communication.

Completion condition: The integrated baseline is controlled, interfaces are defined, risks are managed, and decision-gate evidence is accepted.

1.2 Launch and Lunar Transportation

Includes payload integration, launch-service planning, trans-lunar transportation, lunar-orbit operations, descent and landing, payload separation, surface-delivery verification, and transportation contingencies. Development of a new launch vehicle is excluded.

Completion condition: Required assets are safely delivered, separated, located, and transferred to the appropriate lunar-surface project.

1.3 Robotic Deployment and Site Preparation

Includes robotic commissioning and communications, terrain and hazard surveys, site preparation and grading, utility and equipment placement, infrastructure inspection, repair support, and remote readiness verification.

Completion condition: The selected site satisfies approved readiness criteria before habitat deployment and crew launch.

1.4 Habitat and Crew Readiness

Includes habitat deployment and commissioning, pressure and atmospheric control, life support, water, food, consumables, waste management, medical capability, radiation protection, emergency response, crew communications, and safe-departure capability.

Completion condition: The habitat is verified to support seven crew members for a 45-day nominal mission with at least 60 days of critical provisions.

1.5 Lunar Data Center

Includes computing equipment, data storage and processing, dedicated power distribution, energy storage, heat rejection, communications, cybersecurity, autonomy, monitoring, load management, fault isolation, recovery, and maintenance capability.

Completion condition: Critical computing operates within approved power, thermal, communications, cybersecurity, maintenance, and recovery limits.

1.6 Shared Infrastructure and Resilience

Includes common surface power, shared communications and navigation, common thermal services, logistics, spares, dust mitigation, environmental and hazard monitoring, emergency power, cross-system fault response, contingency operations, and integrated recovery exercises.

Completion condition: Common services support the robotic, habitat, and data-center projects without compromising crew survival or safe departure.

Next authorized artifact

Develop the WBS Dictionary as a separate controlled artifact. Each entry will define scope, exclusions, deliverables, acceptance criteria, dependencies, assumptions, accountable ownership, risks, and applicable decision gates.

Revision record: Version 0.1, 21 September 2026. Initial Level 1 and Level 2 WBS draft derived from Program Charter v1.1; hierarchy diagram added before baseline approval.